



Isothermal (vapour + liquid) equilibria for binary mixtures of diisopropyl ether with (methanol, or ethanol, or 1-butanol): Experimental data, correlations, and predictions

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ABSTRACT

A glass dynamic recirculating still was employed for the measurement of isothermal (vapour + liquid) equilibrium (VLE) data for the binary mixtures of diisopropyl ether (DIPE) + alcohol, viz. (DIPE + methanol), (DIPE + ethanol), and (DIPE + 1-butanol) at $T = (305.15, 315.15, \text{ and } 325.15) \text{ K}$, $T = (313.15, 323.15, \text{ and } 333.15) \text{ K}$ and $T = (318.15, \text{ and } 338.15) \text{ K}$, respectively. The combined standard uncertainties in the reported system pressures, temperatures and phase compositions are $\pm 0.2 \text{ kPa}$, $\pm 0.1 \text{ K}$ and ± 0.003 , respectively. Maximum pressure azeotropes were observed for all isotherms of the (DIPE + methanol) and (DIPE + ethanol) systems. The experimental results were correlated using both the γ - ϕ and ϕ - ϕ approaches. For the correlation of the VLE data with the γ - ϕ approach, the Wilson, NRTL and UNIQUAC G^E models with the truncated two-term virial equation of state (Hayden and O'Connell correlation for second virial coefficient computation) were used. In the ϕ - ϕ correlation approach, the Peng–Robinson equation of state was used with the Wong–Sander mixing rules incorporating the same G^E models used in the γ - ϕ approach. Comparisons between the experimental values and predictions using UNIFAC (Dortmund) and the Predictive Soave–Redlich–Kwong (PSRK) model were performed to test the predictive capabilities of these models for the experimental data measured here. The thermodynamic consistency of the experimental data was checked with the Herington area test.

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1. Introduction

Phase equilibrium studies on oxygenate blends (such as alcohols and ethers) are important for the automotive fuels industry due to the efficacy of these types of additive combinations in enhancing the octane rating of reformulated gasoline (RFG) and reducing harmful emissions from the combustion of traditional fuel mixtures [1]. Studies on alkyl ether oxygenate additives in gasoline blends have been focused mainly on methyl *tert*-butyl ether (MTBE), ethyl *tert*-butyl ether (ETBE) and *tert*-amyl methyl ether (TAME) [2]. However, concerns over environmental persistence and toxicity of some of these ethers have sparked many debates concerning their usage [3–6].

Diisopropyl ether (DIPE) is a branched or tertiary ether regarded as a less polluting and non-toxic alkyl ether. It has been the subject of a number of studies to determine its potential as a possible additive in oxygenate blends [7–10]. In particular, the low water solubility of DIPE [7] eases concerns over groundwater contamination, one of the major drawbacks with the use of MTBE and ETBE [3–6]. Ethanol has the status of a traditionally favoured biomass-based

renewable oxygenate additive for enhancing the octane rating [11] in RFGs, however, the hygroscopic nature of ethanol compromises the homogeneity of the RFG blend. Other alcohols of interest as oxygenate additives or biofuels include methanol [12] and 1-butanol [13]. The blending of a poorly water-soluble oxygenate, such as DIPE, with these alcohols can assist in this regard, by affecting the miscibility behaviour in the (alcohol + water) system, as observed in the liquid–liquid equilibrium studies by Hwang *et al.* [7].

In addition to the role of oxygenate blends in enhancing the combustion characteristics of RFGs, it is also desirable to have a reduction in the emission levels from the combustion of gasoline. The study of physicochemical or phase equilibrium properties such as vapour pressures [14] and (vapour + liquid) equilibrium (VLE) [15] are important to determine the volatility characteristics of the system. In this study, P - x - y VLE measurements were obtained using a dynamic recirculating still at $T = (305.15, 315.15, \text{ and } 325.15) \text{ K}$ for (DIPE + methanol) at $T = (313.15, 323.15, \text{ and } 333.15) \text{ K}$ for (DIPE + ethanol), and, at $T = (318.15, \text{ and } 338.15) \text{ K}$ for (DIPE + 1-butanol). A (cyclohexane + ethanol) test system at $T = 323.15 \text{ K}$ [16] was studied initially to test the accuracy of the measurement procedure in this study.

To our knowledge, there have been no measurements performed for the (DIPE + ethanol) system at temperatures lower than

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333.15 K and in this study measurements were undertaken at $T = (313.15, 323.15, \text{ and } 333.15) \text{ K}$. Measurements have already been undertaken for this system by Lee and Hu [17] over the temperature range of $T = (343 \text{ to } 363) \text{ K}$ and by Kim *et al.* [18] at 333.15 K. For the latter, the technique of headspace gas chromatography (HSGC) was used for the VLE measurements and comparison with the results obtained in this study indicates significant discrepancies between the two data sets. For the (DIPE + methanol) system, two isothermal data sets were available at $T = (320 \text{ and } 330) \text{ K}$ [19] and for the (DIPE + 1-butanol) system, limited low pressure isothermal data are available from literature in the form of a data set at $T = 313.15 \text{ K}$ as determined by Villamañán *et al.* [20].

The experimental results obtained here have been correlated with thermodynamic models using both the γ - ϕ and ϕ - ϕ approaches. For the former, different liquid phase activity coefficient models, viz. Wilson [21], NRTL [22] and UNIQUAC [23], were used together with the truncated two-term virial equation of state with the Hayden and O'Connell (HOC) second virial coefficient correlation [24] for the vapour phase. For the ϕ - ϕ approach, the Peng–Robinson [25] equation of state (EOS) was used with the Wong–Sandler mixing rules [26] incorporating the Wilson, NRTL and UNIQUAC G^E models. The experimental values were also compared to the predictions from the UNIFAC (Dortmund) and Predictive Soave–Redlich–Kwong (PSRK) models [27] to assess the predictive capability of these approaches for the systems under study here.

2. Experimental

2.1. Materials

The chemicals used in this study were cyclohexane (CAS No. 110-82-7), methanol (CAS No. 67-56-1), ethanol (CAS No. 64-17-5), 1-butanol (CAS No. 71-36-3) and diisopropyl ether (CAS No. 108-20-3). The suppliers, stated purities and associated physical properties (refractive indices and densities), together with the literature values [28–31] of the pure components, are shown in table 1. The refractive indices were measured with a digital refractometer (ATAGO, model RX-5000 α , Japan) with an uncertainty of ± 0.0001 . The densities were measured with a vibrating tube densitometer (Anton Paar, model DMA 5000, Austria) with an uncertainty of $\pm 0.00004 \text{ g} \cdot \text{cm}^3$. The chemicals were not purified further and were stored over 0.3 nm molecular sieves supplied by Merck (South Africa). As observed from table 1, there is good agreement between the literature and experimental refractive indices and densities.

2.2. Apparatus and procedure

An all-glass dynamic liquid and vapour condensate recirculating VLE still was employed for all the measurements in this study. This apparatus has been developed in our laboratory and has been presented elsewhere [6,16,32–34]. The equilibrium still has an approximate volumetric capacity of 85 cm^3 . The salient features of the apparatus include an angularly symmetric vacuum-jacketed Cottrell tube, packed equilibrium chamber and the incorporation of agitation in the reboiler and sample traps. The VLE measurement and control system consist of a pressure transmitter (WIKA model P-10, range: (0 to 0.25) MPa, Germany), a Pt-100 temperature sensor (WIKA class A, South Africa), solenoid valves (Clippard model ET-3-24, USA), a ballast tank, a vacuum pump, variable voltage (0 to 240) V transformers (AC/DC Dynamics, South Africa), a personal computer and a data logging unit (Agilent model 34790A, USA) for the pressure and temperature. The temperature control for the isothermal runs ($\pm 0.03 \text{ K}$) was achieved through a variation of

TABLE 1

Suppliers, stated purities, refractive indices (n_D) and densities (ρ) of chemicals used in this study.^a

Component	Supplier	Purity ^b	n_D (298.15 K)		ρ (298.15 K)/ ($\text{g} \cdot \text{cm}^3$)	
			Exp.	Lit.	Exp.	Lit.
Cyclohexane	Merck	>0.98	1.42655 ^c	1.4265 ^d	0.778414 ^c	0.7785 ^d
Methanol	Merck	>0.995	1.32643	1.32645 ^e	0.786624	0.7866 ^e
Ethanol	Merck	>0.995	1.35929	1.35941 ^f	0.784890	0.78493 ^f
1-Butanol	Fluka	>0.995	1.39705	1.39702 ^g	0.805851	0.80589 ^g
Diisopropyl ether	Sigma-Aldrich	>0.99	1.36547	1.3655 ^f	0.718511	0.71854 ^f

^a Standard uncertainties u are $u(\rho) = 0.00004 \text{ g} \cdot \text{cm}^3$ and $u(n_D) = 0.0001$.

^b Supplier mass fraction purity.

^c At $T = 293.15 \text{ K}$.

^d Reference [28].

^e Reference [29].

^f Reference [30].

^g Reference [31].

pressure (with an algorithm for dT/dP) with a pressure controller (Shinko model ACS-13A, USA) and pulse-width modulation of the Clippard solenoid valves. The temperature sensor was calibrated through the use of a hand-held thermometer calibration unit (WIKA model CTH 6500, Germany) in conjunction with a micro calibration bath (WIKA model CTB 9100-165, Germany) and the pressure transmitter was calibrated with a low pressure calibration unit (WIKA model CPH 6000, Germany). The calibration units were supplied by WIKA Instruments (South Africa). The uncertainties arising from the calibration of the temperature and pressure sensors are $\pm 0.06 \text{ K}$ and $\pm 0.03 \text{ kPa}$, respectively.

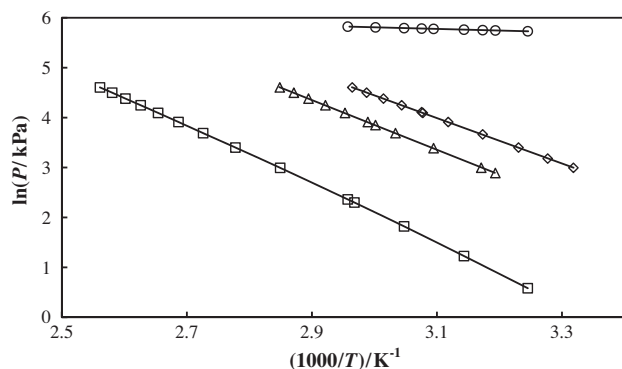
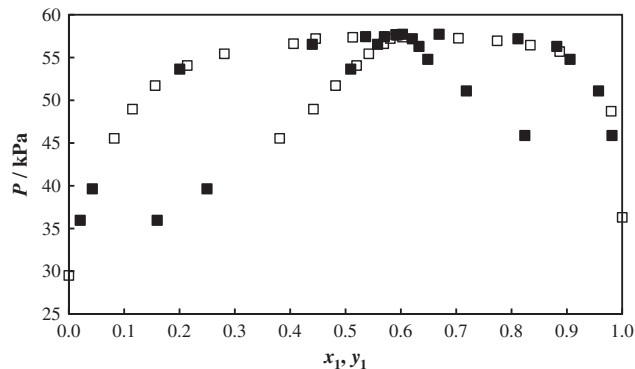
The general procedure for obtaining isothermal VLE data has been described in detail elsewhere [6,16,32–34]. The attainment of the equilibrium condition was monitored closely as a function of heat input into the still, together with the circulation rates and flow patterns of the (vapour + liquid) mixture in the still. After smooth boiling and continuous flow patterns in the apparatus had been established, the time to reach equilibrium was approximately 30 min. The maximum uncertainties associated with the measurement of the plateau regions are $\pm 0.03 \text{ K}$ and $\pm 0.07 \text{ kPa}$ for temperature and pressure, respectively. The combined or overall standard uncertainties (u) in this study were determined in accordance with NIST recommendations outlined by Chirico *et al.* [35]. With regard to the temperature and pressure values reported here, the combined standard uncertainties are determined to be $\pm 0.1 \text{ K}$ and $\pm 0.2 \text{ kPa}$, respectively. Phase composition measurements were obtained with a commercial gas chromatograph (Shimadzu model 2014, Japan) equipped with a thermal conductivity detector (TCD) on an OV-17 packed column (10% silicone, 3 m length, 3.2 mm OD SS) obtained from Trilab Support (South Africa). Dry ultrapure helium (mass fraction purity: 0.99999) supplied by Afrox Gases (South Africa) was used as the carrier gas. Calibration of the TCD was performed with the area ratio method of Raal and Mühlbauer [36]. The injections were performed at least three times for each determination to check repeatability. The combined standard uncertainty in the phase composition analysis in terms of mole fraction for both liquid and vapour phases was determined to be ± 0.003 .

3. Results and discussion

The experimental pure component boiling point temperatures for DIPE, methanol, ethanol and 1-butanol have been compared against literature values obtained from the NIST online databank [37] as shown in table 2 and figure 1. The results of the isothermal VLE test system of (cyclohexane + ethanol) are shown in table 3 and have been plotted in figure 2 against the literature data from

TABLE 2Experimental (T_{exp}) and literature (T_{lit}) boiling point temperatures for diisopropyl ether, methanol, ethanol, and 1-butanol.^a

P/kPa	T_{exp}/K	T_{lit}/K	$(T_{\text{exp}} - T_{\text{lit}})/\text{K}$	P/kPa	T_{exp}/K	T_{lit}/K	$(T_{\text{exp}} - T_{\text{lit}})/\text{K}$
<i>Diisopropyl ether</i> [$\log P/\text{bar} = 3.96649 - (1135.034/(T/\text{K} - 54.92))^b$]							
27.08	305.15	305.27	-0.12	54.08	323.15	323.03	0.12
30.42	308.15	308.09	0.06	57.95	325.15	324.95	0.20
37.15	313.15	313.09	0.06	64.54	328.15	327.98	0.17
39.88	315.15	314.91	0.24	76.91	333.15	333.08	0.07
44.94	318.15	318.03	0.12	90.04	338.15	337.83	0.32
<i>Methanol</i> [$\log P/\text{bar} = 5.20409 - (1581.341/(T/\text{K} - 33.50))^b$]							
19.97	301.37	301.35	0.02	60.94	325.15	325.30	-0.15
24.04	305.15	305.06	0.09	69.96	328.57	328.57	0.00
29.98	309.56	309.61	-0.05	79.92	331.74	331.78	-0.04
39.04	315.15	315.25	-0.10	89.99	334.73	334.71	0.02
50.01	320.72	320.75	-0.03	100.00	337.33	337.37	-0.04
<i>Ethanol</i> [$\log P/\text{bar} = 5.24677 - (1598.673/(T/\text{K} - 46.424))^b$]							
17.99	313.15	313.24	-0.09	59.93	338.66	338.73	-0.07
20.00	315.38	315.30	0.08	69.95	342.28	342.37	-0.09
29.60	323.15	323.23	-0.08	79.91	345.48	345.57	-0.09
40.00	329.64	329.64	0.00	89.92	348.31	348.46	-0.15
47.01	333.15	333.20	-0.05	100.07	351.05	351.14	-0.09
49.99	334.52	334.58	-0.06				
<i>1-Butanol</i> [$\log P/\text{bar} = 4.54607 - (1351.555/(T/\text{K} - 93.34))^b$]							
1.79	308.15	308.10	0.05	40.01	366.84	366.72	0.12
3.41	318.15	318.10	0.05	50.01	372.22	372.18	0.04
6.18	328.15	328.19	-0.04	60.00	376.79	376.81	-0.02
9.97	336.93	336.97	-0.04	69.96	380.79	380.83	-0.04
10.61	338.15	338.17	-0.02	79.92	384.36	384.41	-0.05
19.97	350.98	350.99	-0.01	89.99	387.52	387.68	-0.16
29.98	360.02	359.96	0.06	100.00	390.47	390.64	-0.17

^a Standard uncertainties u are $u(P) = 0.2$ kPa and $u(T) = 0.1$ K.^b From the NIST Chemistry WebBook [37]**FIGURE 1.** Experimental vapour pressures of diisopropyl ether (o), methanol (◇), ethanol (Δ) and 1-butanol (□) compared with literature values (—) [37].**FIGURE 2.** P - x_1 - y_1 diagram for {cyclohexane (1) + ethanol (2)} at $T = 323.15$ K: (■), experimental values; (□), literature values [16].**TABLE 3**Isothermal VLE data for {cyclohexane (1) + ethanol (2)} at $T = 323.15$ K.^a

P/kPa	x_1	y_1	P/kPa	x_1	y_1
35.97	0.021	0.160	57.73	0.669	0.603
39.65	0.043	0.250	57.20	0.812	0.621
53.65	0.200	0.509	56.30	0.882	0.633
56.55	0.440	0.558	54.79	0.906	0.649
57.44	0.536	0.570	51.10	0.957	0.718
57.67	0.591	0.592	45.88	0.982	0.824
57.68	0.593	0.595			

^a Standard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.2$ kPa and $u(x_1) = u(y_1) = 0.003$.

the work of Joseph *et al.* [16]. The experimental P - x - y results for {DIPE (1) + methanol (2)} at $T = (305, 315.15, \text{ and } 325.15) \text{ K}$ are presented in table 4 and plotted in figure 3. The experimental P - x - y values for {DIPE (1) + ethanol (2)} at $T = (313, 323.15, \text{ and } 333.15) \text{ K}$ are presented in table 5 and plotted in figure 4 and the

data for {DIPE (1) + 1-butanol (2)} at $T = (318.15 \text{ and } 338.15) \text{ K}$ are presented in table 6 and plotted in figure 5.

3.1. Comparison with literature data

There is good agreement between the experimental and literature vapour pressure values, as shown in table 2. The verification of the experimental procedure in this study was achieved through the measurement of the (cyclohexane + ethanol) test system and comparison with values determined by Joseph *et al.* [16]. The literature data set was verified as being thermodynamically consistent by Joseph *et al.* Hence there were minimal concerns over the injudicious selection of a suitable test system. Excellent agreement between the two sets of values in figure 2 confirms the reliability of the experimental technique in this study for the measurement of isothermal VLE systems.

TABLE 4

Isothermal VLE data and activity coefficients (γ_i) for {diisopropyl ether (1) + methanol (2)}.^a

P/kPa	x_1	y_1	γ_1	γ_2
<i>T</i> = 305.15 K				
32.91	0.051	0.302	7.30	1.00
37.39	0.107	0.410	5.33	1.02
39.56	0.158	0.459	4.26	1.05
41.19	0.219	0.495	3.44	1.10
42.29	0.310	0.537	2.70	1.18
42.36	0.396	0.545	2.15	1.32
42.43	0.468	0.550	1.84	1.49
42.45	0.527	0.553	1.64	1.67
42.47	0.539	0.556	1.61	1.70
42.49	0.546	0.561	1.61	1.71
42.53	0.571	0.565	1.55	1.79
42.33	0.632	0.548	1.35	2.16
41.95	0.759	0.567	1.15	3.13
41.15	0.829	0.587	1.07	4.14
39.75	0.881	0.619	1.03	5.31
35.51	0.946	0.723	1.00	7.66
<i>T</i> = 315.15 K				
43.91	0.011	0.122	12.56	1.00
49.39	0.037	0.240	8.19	0.99
54.56	0.071	0.336	6.55	0.99
58.19	0.109	0.391	5.27	1.01
62.06	0.200	0.453	3.53	1.08
64.53	0.326	0.489	2.42	1.25
65.62	0.458	0.510	1.83	1.51
65.37	0.507	0.513	1.65	1.65
65.57	0.519	0.516	1.63	1.68
65.68	0.526	0.518	1.62	1.71
64.86	0.587	0.525	1.45	1.91
64.52	0.706	0.541	1.23	2.58
63.96	0.779	0.558	1.14	3.28
61.45	0.859	0.597	1.07	4.51
58.76	0.898	0.629	1.03	5.51
52.51	0.945	0.722	1.00	6.90
<i>T</i> = 325.15 K				
68.86	0.021	0.126	7.36	1.01
75.89	0.045	0.229	6.81	1.00
82.85	0.079	0.320	5.86	1.00
90.96	0.159	0.409	4.05	1.04
96.47	0.300	0.471	2.60	1.19
96.87	0.459	0.490	1.77	1.49
96.83	0.488	0.495	1.69	1.56
96.84	0.501	0.500	1.66	1.59
96.85	0.504	0.509	1.68	1.57
96.94	0.549	0.513	1.55	1.71
96.84	0.627	0.515	1.36	2.06
96.76	0.689	0.516	1.24	2.47
95.75	0.742	0.536	1.18	2.83
93.67	0.809	0.560	1.11	3.55
86.87	0.901	0.627	1.03	5.42
76.95	0.957	0.728	1.00	8.15

^a Standard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.2$ kPa and $u(x_1) = u(y_1) = 0.003$.

The experimental P - x - y isotherms for (DIPE + ethanol) are shown in figure 4. Maximum pressure azeotropes are observed for the systems under study, which is consistent with the work by Lee and Hu [17], who obtained isothermal results for the same two components from $T = (344.17$ to $363.20)$ K. The (DIPE + ethanol) system at 333.15 K has been studied previously by the group of Kim *et al.* [18] through the use of HSGC. In their study, the key experimental variable of equilibrium pressure was not measured directly, but was instead calculated from the vapour phase compositions and thermodynamic relations [18]. An excellent discussion of the applicability of HSGC for VLE measurements is provided by Luis *et al.* [38] who mentioned the necessity for a cautious approach when measuring VLE properties at finite concentrations with HSGC.

A direct comparison between the experimental results for the (DIPE + ethanol) system at $T = 333.15$ K determined in this study

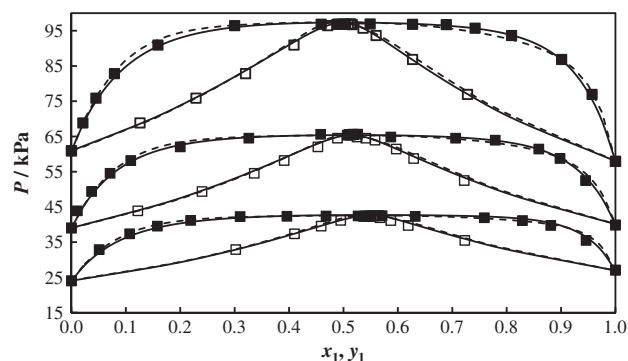
FIGURE 3. P - x_1 - y_1 diagram for {diisopropyl ether (1) + methanol (2)} at (from bottom to top) $T = (305.15, 315.15, \text{ and } 325.15)$ K: (■, □), experimental values; (—), NRTL-HOC; (---), PR-WS-NRTL.

TABLE 5

Isothermal VLE data and activity coefficients (γ_i) for {diisopropyl ether (1) + ethanol (2)}.^a

P/kPa	x_1	y_1	γ_1	γ_2
<i>T</i> = 313.15 K				
26.73	0.063	0.359	4.16	1.01
31.97	0.125	0.480	3.34	1.05
34.90	0.179	0.546	2.88	1.07
36.78	0.223	0.585	2.61	1.09
38.37	0.273	0.611	2.32	1.14
40.06	0.340	0.640	2.03	1.21
41.10	0.420	0.663	1.75	1.32
42.19	0.651	0.691	1.20	2.07
42.22	0.714	0.714	1.14	2.34
42.45	0.726	0.726	1.14	2.36
41.76	0.895	0.821	1.03	3.97
<i>T</i> = 323.15 K				
44.39	0.086	0.367	3.56	1.03
52.56	0.168	0.499	2.91	1.06
57.19	0.259	0.567	2.32	1.12
59.95	0.345	0.605	1.95	1.22
61.26	0.416	0.625	1.70	1.32
62.56	0.568	0.654	1.33	1.69
63.64	0.696	0.698	1.18	2.13
63.52	0.706	0.717	1.19	2.06
63.46	0.779	0.728	1.09	2.64
61.96	0.881	0.789	1.02	3.72
59.51	0.945	0.862	1.00	5.08
<i>T</i> = 333.15 K				
63.85	0.064	0.296	3.92	1.02
72.89	0.125	0.405	3.11	1.05
83.06	0.235	0.518	2.39	1.11
87.47	0.300	0.564	2.14	1.15
89.96	0.379	0.599	1.85	1.22
91.68	0.521	0.623	1.42	1.53
92.73	0.671	0.675	1.21	1.94
92.75	0.684	0.686	1.20	1.96
92.67	0.810	0.702	1.04	3.09
89.87	0.886	0.767	1.01	3.92
85.95	0.937	0.838	1.00	4.74

^a Standard uncertainties u are $u(T) = 0.1$ K, $u(P) = 0.2$ kPa and $u(x_1) = u(y_1) = 0.003$.

with that of Kim *et al.* [18], as shown in figure 6, yields significant discrepancies between the two sets of data. The data of Lee and Hu [17] for the (DIPE + ethanol) system were shown to be thermodynamically consistent (through the point, area and infinite dilution tests) and were utilized as a reference set to effect a comparison between the data of Kim *et al.* and those obtained in this study. Two data sets of Lee and Hu at $T = (344.17$ and $363.20)$ K have been simultaneously regressed with the PR-WS-NRTL model using the same experimental uncertainties in the maximum likelihood approach used by the original authors. The

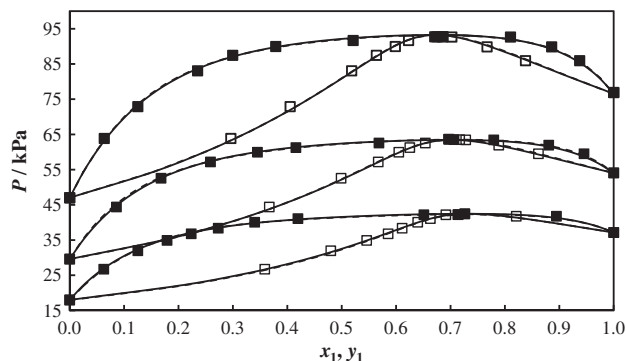


FIGURE 4. P - x_1 - y_1 diagram for {diisopropyl ether (1) + ethanol (2)} at (from bottom to top) $T = (313.15, 323.15, \text{ and } 333.15) \text{ K}$: (■, □), experimental values; (—), NRTL-HOC; (---), PR-WS-NRTL.

TABLE 6

Isothermal VLE data and activity coefficients (γ_i) for {diisopropyl ether (1) + 1-butanol (2)}.^a

P/kPa	x_1	y_1	γ_1	γ_2
$T = 318.15 \text{ K}$				
9.94	0.042	0.620	3.35	1.15
14.86	0.101	0.774	2.59	1.09
26.27	0.258	0.885	2.03	1.18
31.76	0.394	0.920	1.67	1.22
32.74	0.415	0.923	1.63	1.25
33.66	0.447	0.926	1.56	1.31
37.93	0.630	0.946	1.27	1.60
39.06	0.701	0.951	1.18	1.85
42.89	0.895	0.973	1.04	3.20
$T = 338.15 \text{ K}$				
21.95	0.039	0.511	3.33	1.05
37.05	0.131	0.743	2.41	1.02
46.46	0.201	0.799	2.10	1.09
21.95	0.039	0.511	3.33	1.05
37.05	0.131	0.743	2.41	1.02
46.46	0.201	0.799	2.10	1.09
60.64	0.345	0.865	1.72	1.16
63.26	0.380	0.875	1.64	1.18
68.96	0.472	0.896	1.47	1.26
72.57	0.543	0.907	1.36	1.37
74.26	0.581	0.912	1.31	1.44
80.19	0.721	0.932	1.16	1.81
88.91	0.945	0.978	1.02	3.30

^a Standard uncertainties u are $u(T) = 0.1 \text{ K}$, $u(P) = 0.2 \text{ kPa}$ and $u(x_1) = u(y_1) = 0.003$.

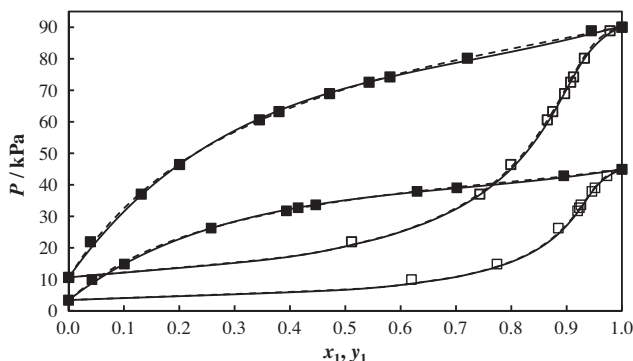


FIGURE 5. P - x_1 - y_1 diagram for {diisopropyl ether (1) + 1-butanol (2)} at (from bottom to top) $T = (318.15, 323.15, \text{ and } 338.15) \text{ K}$: (■, □), experimental values; (—), NRTL-HOC; (---), PR-WS-NRTL.

best-fitted parameters from this regression routine for the PR-WS-NRTL model were employed to predict the P - x - y curve diagram for

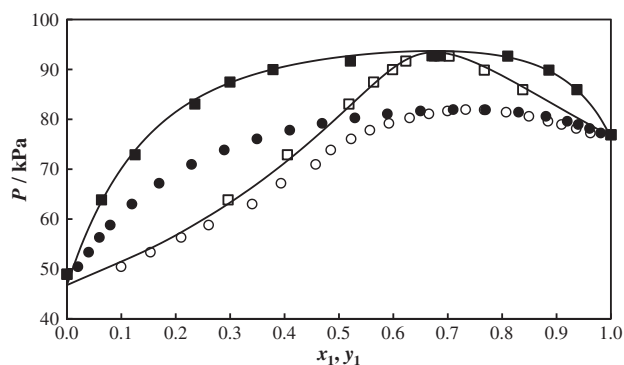


FIGURE 6. Comparison of P - x_1 - y_1 diagrams for {diisopropyl ether (1) + ethanol (2)} at $T = 333.15 \text{ K}$: (■, □), this work; (●, ○), Kim *et al.* [18]; (—), predicted from the best-fitted PR-WS-NRTL parameters obtained from the work of Lee and Hu [17].

the (DIPE + ethanol) system at $T = 333.15 \text{ K}$. The predicted curve and the experimental data sets are shown in figure 6. There is excellent agreement between the P - x - y diagram obtained by regression from the work of Lee and Hu and the results from this study. However, both the P - x and P - y curves from the work of Kim *et al.* are shifted well below the predicted values and the experimental values from this study. These discrepancies are quite significant and a maximum difference in P values of over 10 kPa is obtained near the azeotrope.

The discrepancies described above are consistent with observations from another study by the same group in the form of the work by Oh *et al.* [11] for the isothermal VLE of (ethanol + 2,2,4-trimethylpentane) at $T = 333.15 \text{ K}$. The comparison of the results by Oh *et al.* with those measured with a recirculation VLE still by Hiaki *et al.* [39], exhibited the same trend where the P - x - y curve from the work of Oh *et al.* was shifted below that measured with a VLE recirculation still. Through a favourable comparison of our results with predicted values from other studies, as discussed above, and predictive techniques (shown below), significant arguments have been raised to question the reliability of the data set obtained by Kim *et al.* for the (DIPE + ethanol) system at $T = 333.15 \text{ K}$ in the light of the measurements that have been obtained here.

3.2. Binary VLE calculations and correlations

The γ - ϕ and the ϕ - ϕ approaches were both used for the correlation of the VLE values as a means of comparison of the two methods, as has been performed by other researchers [40] for low pressure VLE data. Comprehensive discussions of both approaches have been provided in past works [36,41].

The equilibrium condition in the γ - ϕ method is represented as follows:

$$y_i P \phi_i = x_i \gamma_i P_i^{\text{sat}}, \quad (1)$$

where y_i is the vapour phase mole fraction, P is the system pressure, x_i is the liquid phase mole fraction, γ_i is the activity coefficient and P_i^{sat} is the saturated vapour pressure for the pure component i .

The term (ϕ_i) is given as follows:

$$\phi_i = \left(\frac{\phi_i^V}{\phi_i^{\text{sat}}} \right) \exp \left[\frac{(V_i^L / \text{cm}^3 \cdot \text{mol}^{-1})}{(R / \text{kPa} \cdot \text{cm}^3 \cdot \text{mol}^{-1} \cdot \text{K}^{-1})(T / \text{K})} (P_i^{\text{sat}} / \text{kPa} - P / \text{kPa}) \right], \quad (2)$$

where ϕ_i^{sat} is the fugacity coefficient for the saturated vapour phase of the pure component, ϕ_i^V is the fugacity coefficient for the vapour phase, R is the universal gas constant, V_i^L is the molar volume of the liquid phase and T is the system temperature. The truncated two-term virial equation of state was employed for the computation of

TABLE 7Physical properties of the pure components in this study.^a

Component	T_c/K	$P_c/$ MPa	$V_c/$ (cm ³ · mol ⁻¹)	Z_c	ω	r/nm	$\mu/$ Debye
Diisopropyl ether	500.05	2.880	386.00	0.267	0.339	0.3906	1.130
Methanol	512.50	8.084	117.00	0.222	0.566	0.1552	1.700
Ethanol	514.00	6.137	168.00	0.241	0.644	0.2259	1.691
1-Butanol	563.10	4.414	273.00	0.258	0.588	0.3225	1.670

^a As retrieved from the Aspen Properties[®] PURE22 data bank.

the pure component and mixture fugacity coefficients. The second virial coefficients were estimated through the method of Hayden and O'Connell [24]. The Rackett correlation, as modified by Spencer and Danner [42], was employed for the estimation of liquid molar volumes. The parameters for the calculation of the liquid molar volumes and second virial coefficients were retrieved from the Aspen Properties[®] PURE22 databank of Aspen Plus[®] Process Simulator, version 7.0 and are shown in table 7. The pure component vapour pressures used in the regression of the VLE values were those obtained through a fit of the experimental vapour pressures measured here to obtain a unique set of Antoine constants (A , B and C):

$$\ln P = A + \frac{B}{(T + C)}, \quad (3)$$

where P is in kPa and T is in K. The constants are shown in table 8.

The activity coefficients were calculated from the experimental P – T – x – y results determined here with equations (1) and (2), and are shown in tables 4 to 6 for (DIPE + methanol), (DIPE + ethanol) and (DIPE + 1-butanol), respectively. In this study, the Wilson [21], NRTL [22] and UNIQUAC [23] liquid activity coefficient models were used for modelling the liquid phase non-ideal behaviour. These have been described in detail elsewhere [36]. The binary interaction parameters for the activity coefficient models (b_{ij}), as used in Aspen Plus[®] Process Simulator, are shown below:

Wilson:

$$A_{ij} = \exp\left(\frac{b_{ij}}{T}\right), \quad b_{ij} = -\left(\frac{\lambda_{ij} - \lambda_{ii}}{R}\right) \text{ where } i \neq j \text{ for } i, j = 1, 2. \quad (4)$$

NRTL:

$$\tau_{ij} = \left(\frac{b_{ij}}{T}\right), \quad b_{ij} = -\left(\frac{g_{ij} - g_{ii}}{R}\right), \quad G_{ij} = \exp(-\alpha_{ij}\tau_{ij}), \quad \alpha_{ij} = \alpha_{ji} \text{ where } i \neq j \text{ for } i, j = 1, 2. \quad (5)$$

UNIQUAC:

$$\tau_{ij} = \exp\left(\frac{b_{ij}}{T}\right), \quad b_{ij} = -\left(\frac{u_{ij} - u_{ii}}{R}\right) \text{ where } i \neq j \text{ for } i, j = 1, 2. \quad (6)$$

The third parameter of the NRTL model, the non-randomness parameter or α_{ij} was fixed at a value of 0.47.

In the ϕ – ϕ method, an equation of state (EOS) is chosen for the representation of both the liquid and vapour phases. In this study,

the Peng–Robinson EOS [25] with the Wong–Sandler mixing rules [26] incorporating the same G^E models in the γ – ϕ approach was used to model the results. In addition to the appropriate choice of mixing rules for mixture properties, the selection of a suitable alpha function is equally crucial and for this purpose the Mathias–Copeman alpha function [43] was used. The Mathias–Copeman alpha function, shown below for subcritical conditions ($T < T_c$), has three adjustable coefficients (c_1 , c_2 and c_3) and is able to provide accurate representation of the vapour pressures of polar components.

$$\alpha(T) = \left[1 + c_1 \left(1 - \sqrt{\frac{T}{T_c}} \right) + c_2 \left(1 - \sqrt{\frac{T}{T_c}} \right)^2 + c_3 \left(1 - \sqrt{\frac{T}{T_c}} \right)^3 \right]^2, \quad (7)$$

The Mathias–Copeman parameters were regressed to the experimental vapour pressures measured in this study. The regressed Mathias–Copeman alpha function parameters are shown in table 8. The pure component critical properties required for the implementation of the PRWS model have been shown in table 7. There are different approaches to the regression of VLE data using the Wong–Sandler mixing rule when incorporating a G^E model [44]. One approach involves the use of G^E parameters regressed from the γ – ϕ method for the G^E model in the EOS mixing rule so that the G^E values calculated from the activity coefficient models in both approaches will match very closely. However, this approach was not used here as it was decided to utilize each approach independently of one another to effect a meaningful comparison. Consequently, there were three parameters correlated to the PRWS model in the form of the asymmetrical binary interaction parameters (b_{ij} , b_{ji}), as described for the G^E models above, and the symmetrical binary interaction parameter (k_{ij}). The non-randomness parameter α_{ij} for the NRTL model was fixed at the same value as that used for the corresponding system in the γ – ϕ method.

The VLE results were regressed through the use of the maximum likelihood objective function (OF), as shown below:

$$OF = \sum_{i=1}^N \left[\frac{(x_i^{exp} - x_i^{calc})^2}{\sigma_x^2} - \frac{(y_i^{exp} - y_i^{calc})^2}{\sigma_y^2} - \frac{(P_i^{exp} - P_i^{calc})^2}{\sigma_P^2} - \frac{(T_i^{exp} - T_i^{calc})^2}{\sigma_T^2} \right], \quad (8)$$

where the superscripts “exp” and “calc” refer to the experimental and calculated quantities, respectively. The experimental variable standard deviations used in this study are $\sigma_x = 0.003$, $\sigma_y = 0.003$, $\sigma_P = 0.09$ kPa and $\sigma_T = 0.08$ K. All thermodynamic calculations have been carried out in Aspen Plus[®] Process Simulator, version 7.0.

The average absolute deviation (AAD) for a calculated quantity is defined as follows:

$$AAD = \frac{1}{N} \left(\sum_{i=1}^N |\Delta Q| \right), \quad (9)$$

where ΔQ is the difference between the experimental and calculated values. The optimized adjustable parameters for the models, together with the deviations for the calculated quantities are shown in table 9. The γ – ϕ method is widely favoured amongst experimenters as being the approach of choice when representing the VLE of non-ideal systems at low pressures [36], where the truncated two-term virial EOS is suitable for representing the vapour phase and the flexibility of the G^E models is able to capture effectively the liquid phase non-ideal behaviour. Indeed, an examination of table 9 attests to this applicability of the method to the systems and conditions of study here. All three models provide good representation of the VLE data for the three different chemical systems. The Wilson and NRTL equations perform equally well in representing the VLE data of all the systems in this study, and the UNIQUAC

TABLE 8Antoine constants (A , B , C) and Mathias–Copeman alpha function coefficients (c_1 , c_2 , c_3) obtained by fitting to experimental pure component vapour pressures.

Component	Antoine constants			Mathias–Copeman coefficients		
	A	B	C	c_1	c_2	c_3
DIPE	14.162111	–2864.844	–41.5397	0.95012	–0.78437	1.8650
Methanol	14.579031	–2566.002	–80.0485	1.3634	–1.5691	2.5779
Ethanol	16.411996	–3536.759	–51.5975	1.1313	1.4648	–3.8562
1-Butanol	14.577264	–2878.370	–102.5146	1.0961	0.59562	0.45161

TABLE 9

Model parameters (b_{12} , b_{21}) and average absolute deviations (AAD) for {diisopropyl ether (1) + methanol (2)}, {diisopropyl ether (1) + ethanol (2)} and {diisopropyl ether (1) + 1-butanol (2)} isothermal VLE data using the γ - ϕ method.

Model	b_{12}/K	b_{21}/K	AAD (x_1)	AAD (y_1)	AAD (P)/kPa	AAD (T)/K
{Diisopropyl ether (1) + methanol (2)}						
$T = 305.15 \text{ K}$						
Wilson	-409.112	-598.608	0.004	0.009	0.07	0.10
NRTL ^a	558.630	424.957	0.004	0.008	0.07	0.10
UNIQUAC	-659.433	59.618	0.003	0.004	0.08	0.11
$T = 315.15 \text{ K}$						
Wilson	-454.641	-518.518	0.002	0.004	0.04	0.08
NRTL ^a	502.438	454.109	0.002	0.003	0.04	0.07
UNIQUAC	-624.341	49.569	0.005	0.004	0.04	0.07
$T = 325.15 \text{ K}$						
Wilson	-426.937	-545.257	0.002	0.005	0.02	0.06
NRTL ^a	520.485	431.817	0.002	0.004	0.02	0.06
UNIQUAC	-654.108	60.590	0.004	0.004	0.03	0.08
{Diisopropyl ether (1) + ethanol (2)}						
$T = 313.15 \text{ K}$						
Wilson	-330.972	-361.739	0.004	0.005	0.05	0.05
NRTL ^a	354.327	325.218	0.003	0.005	0.05	0.05
UNIQUAC	-378.582	68.628	0.003	0.004	0.05	0.06
$T = 323.15 \text{ K}$						
Wilson	-316.723	-368.909	0.003	0.005	0.03	0.05
NRTL ^a	360.630	312.437	0.003	0.005	0.03	0.06
UNIQUAC	-395.432	77.966	0.004	0.006	0.06	0.10
$T = 333.15 \text{ K}$						
Wilson	-304.132	-394.923	0.005	0.007	0.03	0.08
NRTL ^a	385.098	301.256	0.005	0.007	0.03	0.08
UNIQUAC	-421.387	88.376	0.005	0.008	0.04	0.10
{Diisopropyl ether (1) + 1-butanol (2)}						
$T = 318.15 \text{ K}$						
Wilson	-137.696	-437.168	0.002	0.006	0.10	0.07
NRTL ^a	412.071	144.096	0.002	0.006	0.11	0.08
UNIQUAC	-292.224	101.792	0.002	0.006	0.13	0.10
$T = 338.15 \text{ K}$						
Wilson	-133.186	-413.41	0.002	0.003	0.05	0.07
NRTL ^a	392.768	139.179	0.003	0.003	0.05	0.08
UNIQUAC	-287.257	103.107	0.004	0.004	0.06	0.10

^a $\alpha_{ij} = 0.47$.

equation is only marginally inferior to the other two models, especially for the (DIPE + ethanol) and (DIPE + 1-butanol) systems.

In the ϕ - ϕ method, the use of the Wong–Sandler mixing rule with the PR EOS allows the incorporation of G^E models, traditionally used in the γ - ϕ approach, to improve the representation of non-ideal systems containing polar condensable components. It achieves this by creating a composition dependence of the EOS mixing rule in the same mould as the G^E model in γ - ϕ approach and thereby provides a basis for comparison between the γ - ϕ and ϕ - ϕ methods. The fits of the PR-WS- G^E models to the experimental data sets are shown in table 10 and the PR-WS-NRTL and the NRTL-HOC approaches are compared against each other in figures 3 to 5 for the (DIPE + methanol), (DIPE + ethanol), and (DIPE + 1-butanol) systems, respectively. The overall quality of the fit of the same G^E models in γ - ϕ the approach is largely unaffected by their implementation in the ϕ - ϕ method. The latter method is able to represent the data as well as the former, especially for the (DIPE + ethanol) and (DIPE + 1-butanol) systems. Hence both the γ - ϕ and the ϕ - ϕ approaches perform quite well overall with the relevant models used here in this study.

3.3. Azeotropic analysis

The determination of the experimental azeotropic conditions i.e. $x_1^{AZ,exp}$ and $P^{AZ,exp}$ in this study for the (DIPE + methanol) and (DIPE + ethanol) systems were achieved through numerical

smoothing of the (x_1 - y_1) vs. x_1 and P - x_1 curves, which can produce better results than using correlative procedures, as described by Pavlíček and Wichterle [45]. The azeotropic compositions for each isotherm are shown in table 11.

The empirical correlation developed by Lee and Hu [17] for the (DIPE + ethanol) azeotrope is shown below:

$$x_a^{AZ} = \frac{1}{1 + m^{1.23}} \quad (10)$$

and

$$m = \sqrt{\frac{P_a^{AZ} - P_a^s}{P_b^{AZ} - P_b^s}} \quad (11)$$

where P_a^s and P_b^s are the saturated vapour pressures of the pure components and components “a” and “b” are defined in a manner so as to make m greater than unity.

The azeotropic results obtained in this study are comparable with the values calculated from the above correlation, as shown in table 11. The azeotropic conditions determined by Kim *et al.* [18] at $T = 333.15 \text{ K}$ for (DIPE + ethanol) are also shown in table 11 for comparison. The significant difference between the azeotropic conditions obtained here and that of Kim *et al.* (about 0.1 mole fraction for $x_1^{AZ,exp}$ and 10 kPa for $P^{AZ,exp}$) is consistent with the general lack of agreement between the two sets of experimental results discussed above.

TABLE 10

Model parameters (b_{12} , b_{21} and k_{12}) and average absolute deviations (AAD) for {diisopropyl ether (1) + methanol (2)}, {diisopropyl ether (1) + ethanol (2)}, and {diisopropyl ether (1) + 1-butanol (2)} with the ϕ - ϕ method.

Model	b_{12}/K	b_{21}/K	k_{12}	AAD (x_1)	AAD (y_1)	AAD (P)/kPa	AAD (T)/K
{Diisopropyl ether (1) + methanol (2)}							
$T = 305.15\text{ K}$							
Wilson	−251.050	−1313.48	0.3888	0.006	0.017	0.06	0.09
NRTL ^a	854.917	239.972	0.4426	0.004	0.015	0.05	0.07
UNIQUAC	−916.477	102.288	0.3578	0.003	0.008	0.07	0.10
$T = 315.15\text{ K}$							
Wilson	−286.366	−969.526	0.3873	0.005	0.009	0.07	0.14
NRTL ^a	784.302	265.456	0.4303	0.004	0.007	0.06	0.12
UNIQUAC	−841.746	36.2334	0.2263	0.003	0.003	0.05	0.08
$T = 325.15\text{ K}$							
Wilson	−271.034	−1153.48	0.3913	0.005	0.010	0.05	0.14
NRTL ^a	857.527	254.498	0.4385	0.003	0.009	0.04	0.12
UNIQUAC	−900.075	80.4312	0.3101	0.002	0.004	0.03	0.08
{Diisopropyl ether (1) + ethanol (2)}							
$T = 313.15\text{ K}$							
Wilson	−152.172	−497.555	0.2986	0.004	0.006	0.06	0.07
NRTL ^a	464.515	176.618	0.2882	0.004	0.006	0.06	0.07
UNIQUAC	−476.133	32.6789	−0.0270	0.003	0.005	0.05	0.06
$T = 323.15\text{ K}$							
Wilson	−280.617	−563.412	0.1934	0.004	0.005	0.02	0.04
NRTL ^a	538.118	335.915	0.1647	0.003	0.005	0.02	0.04
UNIQUAC	−628.673	−131.961	−0.7491	0.003	0.005	0.03	0.05
$T = 333.15\text{ K}$							
Wilson	−171.046	−616.717	0.2667	0.005	0.007	0.03	0.08
NRTL ^a	568.426	208.247	0.2514	0.005	0.007	0.03	0.08
UNIQUAC	−545.809	40.493	−0.0496	0.005	0.007	0.04	0.09
{Diisopropyl ether (1) + 1-butanol (2)}							
$T = 318.15\text{ K}$							
Wilson	−217.431	−687.830	−0.0721	0.002	0.005	0.09	0.06
NRTL ^a	644.946	290.148	−0.1155	0.002	0.005	0.09	0.07
UNIQUAC	−369.432	97.7820	−0.1251	0.002	0.006	0.13	0.10
$T = 338.15\text{ K}$							
Wilson	−251.309	−766.902	−0.1143	0.002	0.003	0.04	0.05
NRTL ^a	700.498	325.481	−0.1475	0.002	0.003	0.04	0.06
UNIQUAC	−385.626	101.853	−0.1333	0.003	0.003	0.06	0.09

^a $\alpha_{ij} = 0.47$.

TABLE 11

Experimental ^a, correlated ^b and predicted ^{c,d} azeotropic compositions $\{x_1^{AZ,exp}, x_1^{AZ,corr}, x_1^{AZ,pred}\}$ and pressures $\{P^{AZ,exp}, P^{AZ,pred}\}$ for {diisopropyl ether (1) + methanol (2)} and {diisopropyl ether (1) + ethanol (2)}.

T/K	$x_1^{AZ,exp}$	$x_1^{AZ,corr}$	$x_1^{AZ,pred}$		$P^{AZ,exp}/\text{kPa}$	$P^{AZ,pred}/\text{kPa}$	
			UNIFAC	PSRK		UNIFAC	PSRK
{Diisopropyl ether (1) + methanol (2)}							
305.15	0.557	1	0.537	0.541	42.52	40.52	42.39
315.15	0.518 ^a		0.511	0.519	65.55	62.27	64.97
325.15	0.502		0.486	0.499	96.84	92.26	96.85
{Diisopropyl ether (1) + ethanol (2)}							
313.15	0.724	0.715	0.741	0.743	42.55	42.56	42.43
323.15	0.698	0.682	0.707	0.713	63.83	63.54	63.41
333.15	0.676; 0.768 ^e	0.652	0.674	0.682	92.74; 81.23 ^e	92.50	92.52

^a This work.

^b Correlation of Lee and Hu [17].

^c UNIFAC (Dortmund).

^d PSRK.

^e Kim et al. [18].

3.4. VLE prediction

For the purposes of comparison, two different predictive approaches were employed here in the form of the modified UNIFAC (Dortmund) [46] and the PSRK [47] models. Both models are group contribution models with the former being a low pressure predic-

tive excess Gibbs energy (G^E) model and the latter being a predictive equation of state for use over wider pressure and temperature ranges. An excellent review of both methods is provided by Gmehling [27].

Both predictive techniques were employed with the Dortmund Data Bank software [48]. The fits of these two models to the

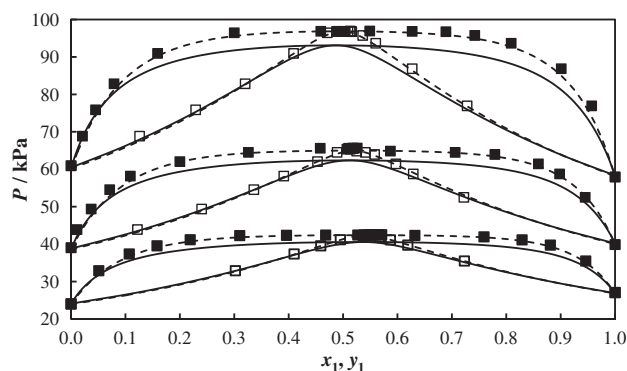


FIGURE 7. Comparison of experimental and predicted P - x_1 - y_1 diagrams for {diisopropyl ether (1) + methanol (2)} with predicted curves at (from bottom to top) $T = (305.15, 315.15, \text{ and } 325.15) \text{ K}$: (■, □), experimental values; (---), UNIFAC (Dortmund); (—), PRSK.

(DIPE + methanol), (DIPE + ethanol) and (DIPE + 1-butanol) experimental values are shown in figures 7 to 9, respectively. There is excellent agreement with both models for the {DIPE + (ethanol or 1-butanol)} systems, however, for the (DIPE + methanol) system, a significant difference in the predictive capabilities of the two models is apparent. The UNIFAC (Dortmund) model is not able to match the experimental P - x_1 data for the (DIPE + methanol) system, which is achieved quite well by the PSRK model. The UNIFAC approach underpredicts the equilibrium pressures for the liquid phase on both sides of the azeotrope. This observation indicates that a judicious approach should be employed when a predictive method is to be employed for the prediction of a VLE data set in the absence of experimental data. The azeotropic compositions predicted by the UNIFAC (Dortmund) and PSRK models are also shown in table 11. There is good agreement between the experimental azeotrope conditions and that predicted by the PSRK model for both the (DIPE + methanol) and (DIPE + ethanol) systems. However, the UNIFAC (Dortmund) underpredicts the azeotropic pressures for the (DIPE + methanol) isotherms, as mentioned above. The overall good agreement of the predictive methods with the experimental values augers well for the predictive capabilities of UNIFAC (Dortmund) and especially the PSRK model for low pressure VLE results with polar components.

3.5. Thermodynamic consistency testing

The validation of the P - T - x - y VLE data measured here was achieved with the Herington area test [49] for thermodynamic

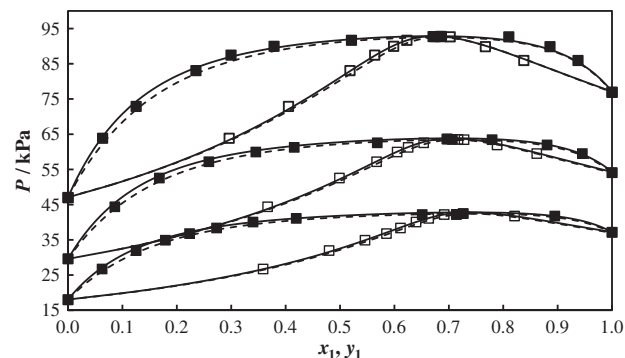


FIGURE 8. Comparison of experimental and predicted P - x_1 - y_1 diagrams for {diisopropyl ether (1) + ethanol (2)} with predicted curves at (from bottom to top) $T = (313.15, 323.15, \text{ and } 333.15) \text{ K}$: (■, □), experimental values; (---), UNIFAC (Dortmund); (—), PRSK.

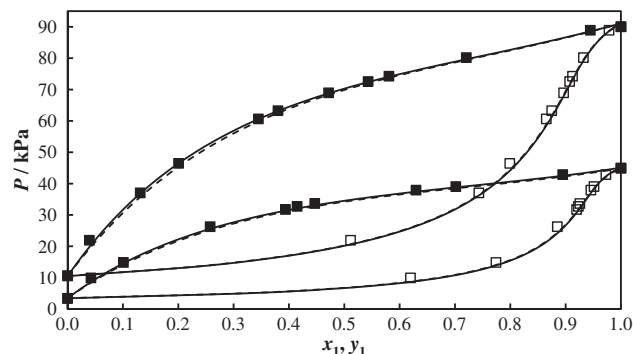


FIGURE 9. Comparison of experimental and predicted P - x_1 - y_1 diagrams for {diisopropyl ether (1) + 1-butanol (2)} with predicted curves at (from bottom to top) $T = (318.15, \text{ and } 338.15) \text{ K}$: (■, □), experimental values; (---), UNIFAC (Dortmund); (—), PRSK.

TABLE 12

Results of the Herington area test for thermodynamic consistency.^a

System	D
{Diisopropyl ether (1) + methanol (2)}	
305.15	2.9
315.15	2.5
325.15	1.5
{Diisopropyl ether (1) + ethanol (2)}	
313.15	3.6
323.15	4.3
333.15	3.9
{Diisopropyl ether (1) + 1-butanol (2)}	
318.15	4.6
338.15	3.5

^a Criterion for passing the test: $D < 5$.

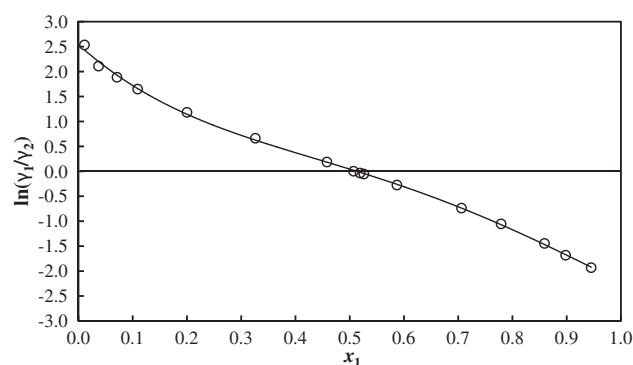


FIGURE 10. Herington area test for the {diisopropyl ether (1) + methanol (2)} system at $T = 315.15 \text{ K}$.

consistency as discussed by Wisniak [50]. The data were considered as being consistent on the basis of D values for the isothermal data case:

$$D = 100 \left| \frac{(A - B)}{(A + B)} \right|, \quad (12)$$

where A and B are the areas above and below the zero line of a plot of $\ln(v_1/v_2)$ versus x_1 , respectively. In order to pass the test, and examination of D is required such that $D < 5$. These have been calculated through the use of polynomial curve fitting (with the polynomial order varied between 2 to 4) and numerical integration with recursive adaptive Simpson quadrature in MATLAB® V.7.9.0. (see table

12). An example of the Herington area test is shown in figure 10, as applied to the (DIPE + methanol) system at $T = 315.15$ K. The results from table 12 indicate that all the VLE data sets measured in this study satisfy the requirements to be considered as thermodynamically consistent by the Herington area test.

4. Conclusions

In this study, isothermal VLE data have been determined for (DIPE + methanol) at $T = (305.15, 315.15, \text{ and } 325.15)$ K, (DIPE + ethanol) at $T = (313.15, 323.15, \text{ and } 333.15)$ K and (DIPE + 1-butanol) at $T = (318.15 \text{ and } 338.15)$ K using a dynamic recirculating still. The results have been correlated with the Wilson, NRTL and UNIQUAC models in the γ - ϕ approach and the Peng–Robinson equation of state and Wong–Sandler mixing rules with the Wilson, NRTL and UNIQUAC G^E models in the ϕ - ϕ approach. Both approaches provided reasonably good fits to the values measured. The predictive approaches of UNIFAC (Dortmund) and PSRK also performed quite well in predicting the P - x - y curves for the systems measured, except for the (DIPE + methanol) system for which UNIFAC (Dortmund) did not perform as well as the PSRK model. The experimental results were subjected to the Herington area test, which provided confirmation of the thermodynamic consistency of the experimental values determined here.

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